

Arca

AI agents for flood rescue

No call for help should go unheard.

Team Orbital · Open Agent Hackathon 2026

Photo: rescuers in Laranjal, Pelotas, 10 May 2024 · Michel Corvello / Prefeitura de Pelotas · public domain

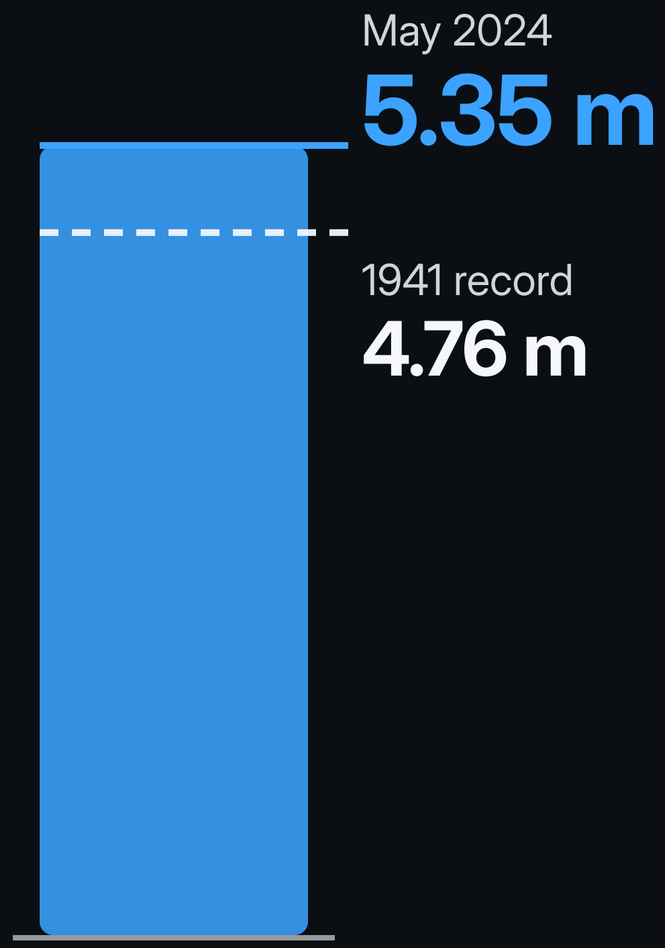
May 2024: the worst flood in Rio Grande do Sul's history.

185

lives lost

2.4 million
people affected

478 of 497
municipalities hit



Guaíba river, Porto Alegre
59 cm above the 1941 record

Defesa Civil RS (final bulletin, 19 Aug 2025); G1 / Serviço Geológico do Brasil

Calls came from everywhere.

77,000+

people rescued

Dozens of WhatsApp groups.
No shared picture.

For civil-defence dispatch rooms and volunteer rescue coordinators.



Defesa Civil RS (30 May 2024); Exame; Agência Lupa · Photo: Laranjal, Pelotas, 10 May 2024 · M. Corvello / Pref. de Pelotas · public domain

A strong El Niño is coming again.

83%

chance of a strong-to-very-strong El Niño through early 2027



The next flood season will need every call answered, fast.

NOAA Climate Prediction Center, ENSO discussion, 8 Oct 2026

One call for help. The agents read it.

WhatsApp · 6 May 2024, 12:00 · replay

“Help!! We’re on the second floor at Voluntários da Pátria 1067, Porto Alegre: me, my bedridden mother and my 8-year-old son. The water won’t stop rising.”

Original, as written:

“socorro!! estamos no segundo andar na rua voluntários da pátria 1067, porto alegre, eu, minha mãe acamada e meu filho de 8 anos, a água não para de subir”

Case card · produced live by the agents

P1 medical · top priority

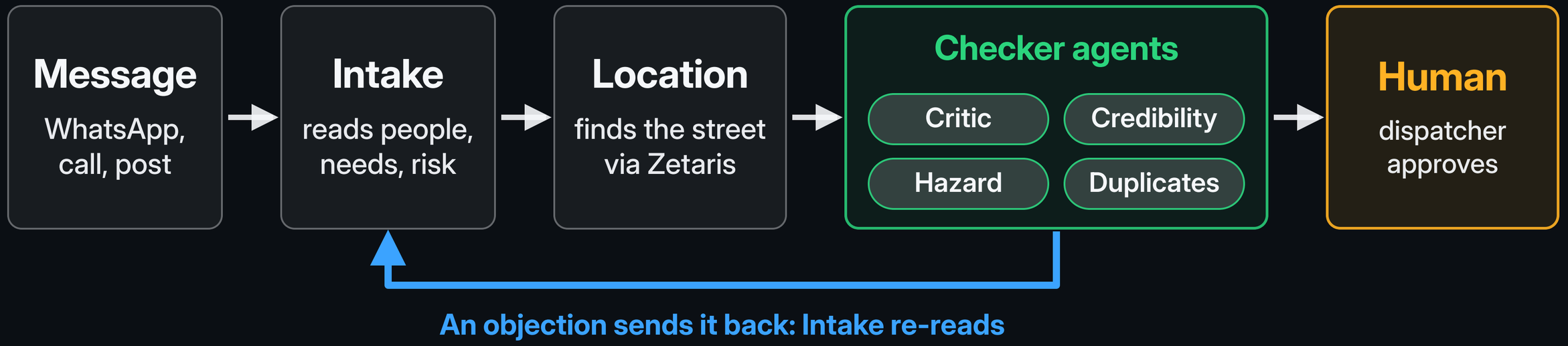
Voluntários da Pátria 1067
✓ address verified in Zetaris

3 people
me · bedridden mother · son, 8

Water ≥ 1.2 m
→ ready to dispatch, pending a human

Live capture, English UI (values redrawn for legibility) · the message is a synthetic replay case written for the demo

Agents check each other's work, live.



Every model call carries a Nemotron receipt.

MODEL	`nvidia/nemotron-3.5-super-vl-120b-a12b`
RECEIPT	`rcpt_req_7ngfp01b · Intake · 5.3 s`

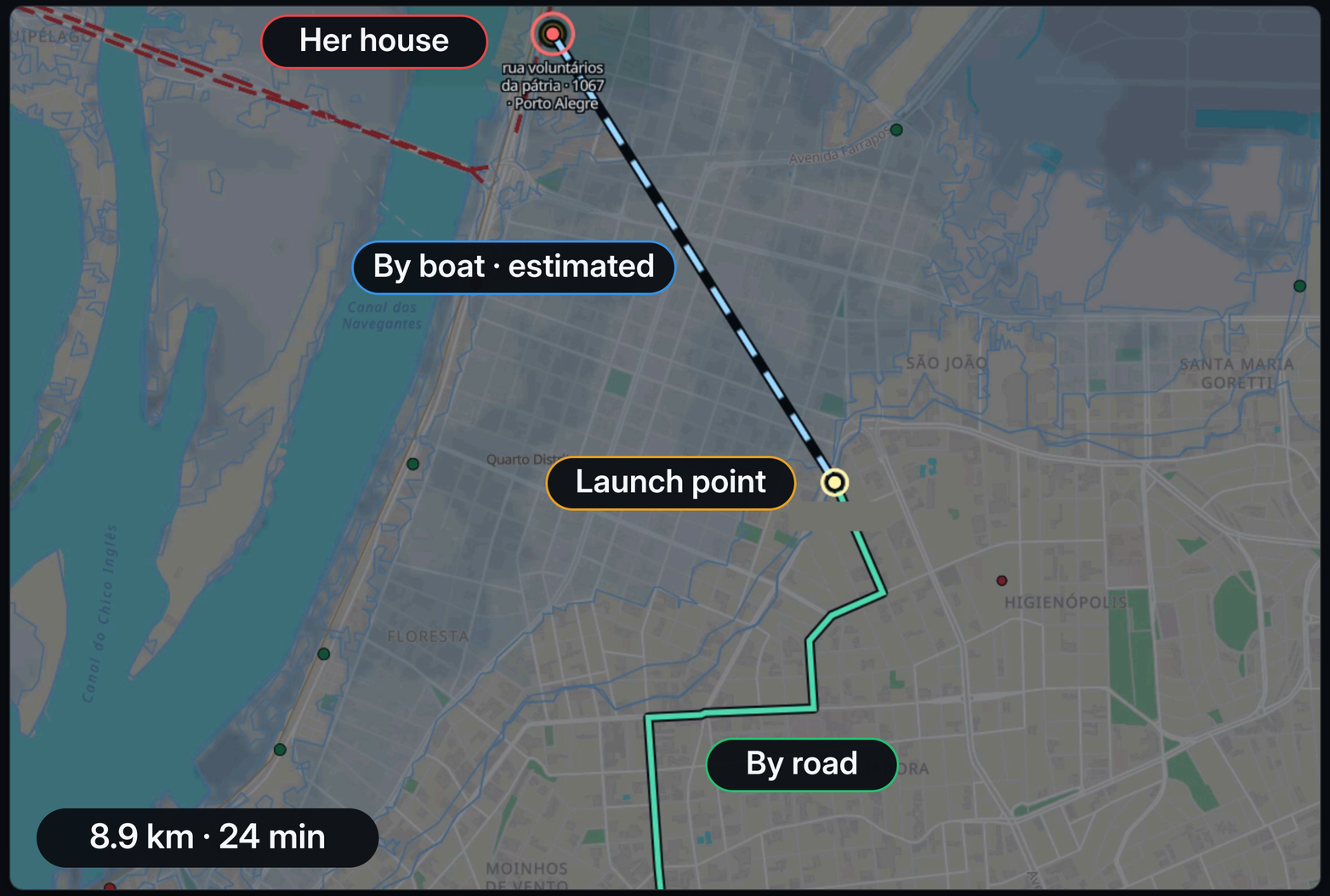
Each agent quotes its evidence.

"minha mãe acamada"

✓ my mother is bedridden → medical

Live run (Oct 2026) on the 6 May 2024 replay · values from the Intake agent's record · checkers run when a case needs them

The agents plan the rescue. A person approves it.



1 The fleet agent picks the boat

Aluminium boat
24 min · 6 seats · 1 trip

2 A person ticks the checks

✓ 5 safety checks

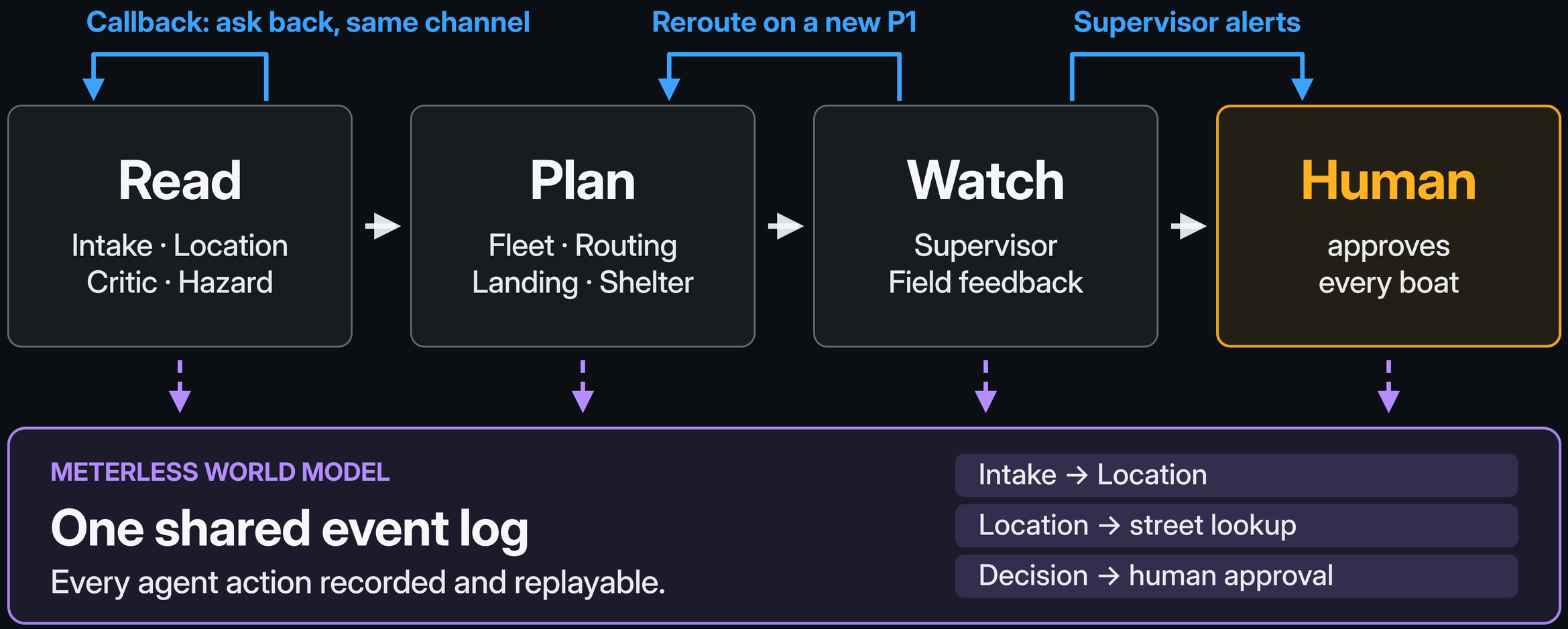
3 Only then: dispatch

Dispatch
10 s to undo

Simulated fleet & GPS.
Water leg is an estimate.

How the agents work together.

28 agents and loops · 19 model agents + 9 operational loops



Counted from the code's agent roster: 19 model agents + 9 operational loops · event log entries from the live run

Built safety-first. Tested hard.

29/30

live requests triaged correctly

11 s average

42/42

red-team attacks stopped

offline attack set

94/100

safe recommendations on never-seen blind requests

run once, live

When unsure, the agents ask again instead of guessing —
and no boat moves without a human.

Blind sets (never seen, run once live, commit 6ba4233): set A 34/50 correct, set B 29/50 correct, 3 unsafe sends each = 94 of 100 safe.
Dev set is ours: 29/30 live, was 19/30; mean 11.3 s, was 46.7 s. Red team 42/42 offline. The blind gap is our next work.

Live and offline runs, 27 Oct 2026 · details in the README evaluation table

Built on three sponsor technologies.

Zetaris knows the ground · Nemotron reads and reasons · Nightshift makes every call accountable · the World Model makes "no boat without a human" a rule

NVIDIA

NVIDIA Nemotron

The model behind every agent decision

METERLESS

Meterless Nightshift + World Model

One gateway for every call; one shared event log

ZETARIS

Zetaris

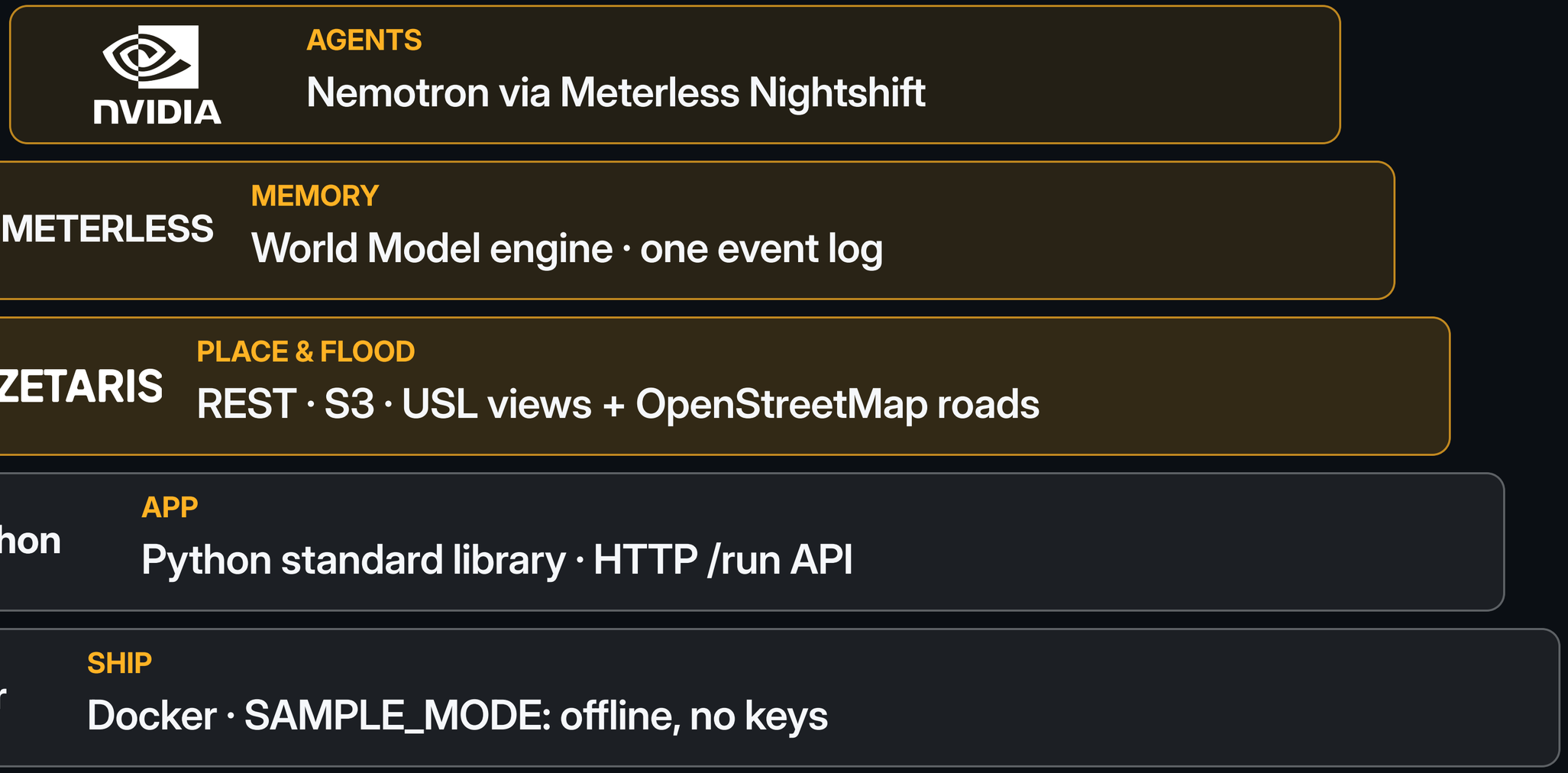
Streets, landmarks and flood data in one query

Where each one sits in the live pipeline

Message → Zetaris finds the place → Nemotron agents via Nightshift → World Model logs every step → Human approves

Logos: official brand assets (NVIDIA, Zetaris), shown monochrome · Meterless shown as a wordmark

A small stack. Every decision replayable.



Architecture and setup: repository README · no keys or tokens are shown or needed for SAMPLE_MODE

What comes next.

1

WhatsApp Business + the 193 line

requests flow into the same queue

2

Live river gauges

Guaíba, Jacuí, Taquari levels via Zetaris

3

Defesa Civil pilot

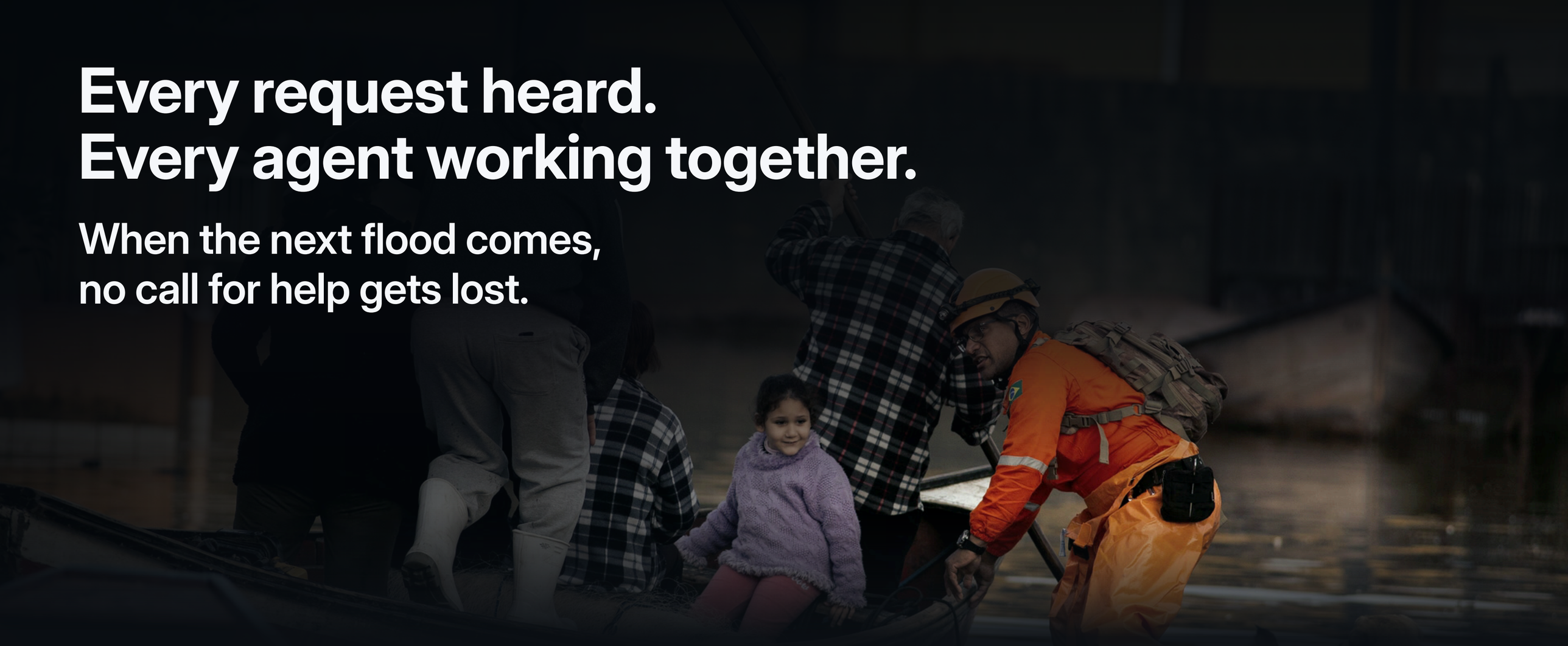
before the next El Niño season

Before any real use: close the gap on never-seen tests.

Roadmap: repository README · call transcripts are text input today; no speech recognition is built

Every request heard.
Every agent working together.

When the next flood comes,
no call for help gets lost.



Arca

Help us pilot it with Defesa Civil before the next El Niño.

github.com/LamboKiller/rescue-triage-desk

Team Orbital · code, evaluations and this deck, open source

Photo: Colônia Z-3, Pelotas, 31 May 2024 · G. Vara / Pref. de Pelotas · public domain

Appendix · Sources & credits

Numbers

185 deaths · 478 of 497 municipalities · 2,398,255 affected
Defesa Civil RS final bulletin, 19 Aug 2025 (sosenchentes.rs.gov.br); IPEA

Over 77,000 rescued (77,729 on 30 May 2024)
Defesa Civil RS bulletin, 30 May 2024

Guaíba 5.35 m (5 May 2024); 1941 record 4.76 m
G1; Serviço Geológico do Brasil

Channels: Exame; Agência Lupa

>83 % strong-to-very-strong El Niño, Jan–Mar 2027
NOAA CPC ENSO discussion, 8 Oct 2026

Dev 29/30 (was 19/30), 11.3 s (was 46.7 s): live, 27 Oct 2026

Red team 42/42: offline attack set

Blind A 34/50, 3 unsafe · Blind B 29/50, 3 unsafe (94/100 safe)
live, run once, commit 6ba4233

28 agents and loops = 19 model agents + 9 operational loops
agents_roster.json

Photographs (Wikimedia Commons)

S1: Resgates, bairro Laranjal (53712679145)
Michel Corvello / Prefeitura de Pelotas · public domain

S3: Resgates, bairro Laranjal (53712420573)
Michel Corvello / Prefeitura de Pelotas · public domain

S11: Missão humanitária, Colônia Z-3 (53761203109)
Gustavo Vara / Prefeitura de Pelotas · public domain

Full URLs: CREDITS.md

Product frames: live captures, English UI

Map: © OpenStreetMap contributors · Protomaps · MapLibre

Logos: NVIDIA, Zetaris official assets

Typeface: Inter (OFL)

Team Orbital · all work built during the build window

Replay of 6 May 2024. Fleet and GPS simulated. Lives saved shown only as potential.